ALGORITHMS AND DATA STRUCTURES

Practice Exam Paper (2 hours)

Module(s): 25COP923

Example paper for revision

Total Marks: 100

Question 1 **[10 marks]**

Definitions

Provide informal definitions of the concepts below.

(a) What is a hash table? Explain how it achieves fast lookup on average. **[5]**

(b) What is a greedy algorithm? State one property that must hold for a greedy approach to guarantee an optimal solution. **[5]**

Question 2 **[15 marks]**

Algorithm Analysis

What is the worst-case time complexity of the following algorithm in big-O notation? What task is performed by the algorithm? How is that task achieved? (Answer each in the corresponding sub-question.)

```
Procedure(A, n, target)
low <- 1, high <- n
while low <= high do
mid <- floor((low + high) / 2)
if A[mid] = target then
return mid
else if A[mid] < target then
low <- mid + 1
else
high <- mid - 1
return -1
```

(a) Complexity? **[5]**

(b) Task performed? **[5]**

(c) How achieved? **[5]**

Question 3 **[20 marks]**

Data Structures

Consider a queue (FIFO) implemented as a singly linked list with head (front) and tail (back) pointers, where enqueue adds to the tail and dequeue removes from the head.

(a) Starting from an empty queue, perform the operations: enqueue(3), enqueue(7), enqueue(1), dequeue(), enqueue(9), dequeue(). State the final contents of the queue (front to back) and the value returned by each dequeue call. **[8]**

(b) Suppose each node has fields 'value' and 'next', and the queue object Q tracks 'head' and 'tail'. Write pseudocode for Enqueue(Q, value). **[6]**

(c) Write pseudocode for Dequeue(Q), which removes and returns the value at the front of the queue. **[6]**

Question 4 **[35 marks]**

Graphs

An emergency response coordinator manages a network of towns connected by roads. Each road has an associated travel time. In the event of a flood, the coordinator needs to know the minimum time required to send aid from the central depot town to every other town in the network.

(a) Explain how this problem can be modelled as a graph problem. **[10]**

(b) Design an algorithm that computes these minimum travel times. Describe it in words, give high-level pseudocode, and state which known algorithm it is based on. **[15]**

(c) Discuss a suitable representation of the graph and an implementation of your algorithm. Mention any auxiliary data structures and estimate the running time. **[10]**

Question 5 **[20 marks]**

Minimum Spanning Trees

Let G = (V, E) be a connected, weighted, undirected graph.

(a) Define what a spanning tree of G is, and what a minimum spanning tree (MST) of G is. **[5]**

(b) State the problem solved by Kruskal's algorithm, giving its input and output. **[5]**

(c) Run the simple (non-optimised) version of Kruskal's algorithm on the graph with vertices {P, Q, R, S, T} and edges P-Q (3), P-R (1), Q-R (4), Q-S (6), R-S (2), R-T (5), S-T (3). List the edges in sorted order, state for each whether it is taken, and give the total weight of the resulting minimum spanning tree. **[10]**